

DOSSIER DE CANDIDATURE CONCOURS DOCTORAL 2025

NOM : Dutel

PRENOM : Jordan

TYPE D'ALLOCATION DEMANDE : 1- UCBL1 ☒ 2-Dons et Legs CLB ☐* 3-Indifférencié ☐

AVEZ-VOUS POSTULE A D'AUTRES FINANCEMENTS ? FRM ☐ ITMO ☐

Autres (préciser) ☒

* La bourse Dons et Legs implique des présentations régulières devant les financeurs et une capacité à vulgariser son sujet pour parler au grand public.

Pour que le dossier soit étudié, il doit être transmis complet en un seul PDF lors du dépôt de candidature par l'étudiant sur ADUM via ce lien <https://adum.fr/candidature/>.

Outre cette fiche, le dossier de candidature doit être en anglais

Voici ce que le PDF doit contenir :

1. Cette Fiche remplie (nom, prénom type d'allocation) en première page du dossier

2. Le Curriculum Vitae de la candidate/du candidat

Dans lequel les critères académiques seront précisés et détaillés. Donnez toutes les informations en votre possession concernant le M1 et le M2, ou équivalent, (notes déjà connues, équivalence éventuelle, rang, etc.), ainsi que la licence/bachelor ou équivalence et le baccalauréat ou équivalence. Si passage en Session 2 faire apparaître obligatoirement (car pas de classement en session 1 si session 2). **Fournir tous les relevés de notes + le rang en votre possession.**

3. Le curriculum vitae de la directrice/du directeur de thèse

Avec la liste des publications les plus significatives sur les 5 dernières années de **l'équipe d'accueil**, avec la place des étudiantes/étudiants en gras et de la directrice/du directeur de thèse soulignés. **Seul.es les directeur.ices et co-directeur.ices avec HDR peuvent figurer dans le dossier.**

4. Le titre et une description du projet de la thèse (5 pages maximum figures, annexes et bibliographies comprises)

5. Le calendrier de réalisation du projet sur 3 ans (1/2 page)

6. Le type d'allocation et si candidature à d'autres financements (demandé en début de page)

7. Un résumé vulgarisé du projet d'une demi page.

Si le choix 2 ou le choix 3 a été sélectionné dans le type d'allocation, un résumé vulgarisé indiquant l'intérêt sociétal du sujet sera transmis aux financeurs. Si le choix 1 est sélectionné, il n'est pas demandé de fournir ce résumé.

Le dossier de notation est à pré-remplir et à renvoyer par mail à la gestionnaire de l'école doctorale, ne surtout pas l'ajouter au PDF : Juliette.MAHE-BOUGUENNEC@lyon.unicancer.fr

Aspiring bioinformatician with a strong background in molecular biology and immunology, seeking to deepen expertise in multi-omics, data integration, and translational cancer research. Motivated to contribute to high-impact scientific projects and pursue a PhD in systems biology or immuno-oncology.

SKILLS

Bioinformatics	Transcriptomics (bulk and spatial), Seurat, Harmony, Nextflow, R, Python, Multi-omics Integration
Data Science	Dimensionality Reduction, Clustering, Differential Expression, Feature Selection, Statistical Modeling
Technical Tools	Git, Anaconda, LaTeX, Shell scripting, High Performance Computing (HPC)
Languages	French (native), English (fluent), Spanish and Italian (notions)

EXPERIENCE

Master 2 Bioinformatics Intern – Spatial Transcriptomic Analysis
CRCL – Centre de Recherche en Cancérologie de Lyon

February 2025 — Present
Lyon, France

- Analyzed spatial transcriptomics data (10X Visium) from 15 triple-negative breast cancer (MpBC) patients.
- Integrated expert-annotated phenotypic cell types to characterize tumor heterogeneity.
- Developed and optimized normalization, feature selection, and batch effect correction pipelines (Seurat, Harmony).
- Identified spatially enriched markers and distinct transcriptional profiles among tumor subtypes.
- Contributed to ongoing research on rare breast cancer evolution and potential PhD project.

Master 1 Bioinformatics Intern – Multi-omics Analysis
CLB – Centre Léon Bérard

April 2024 — August 2024
Lyon, France

- Performed integrative analysis of genomic, transcriptomic, and proteomic data from the Profiler cancer cohort.
- Conducted differential expression and pathway enrichment to identify molecular signatures associated with clinical subtypes.
- Implemented reproducible pipelines in R and Nextflow for data preprocessing and normalization.
- Applied MOFA to identify latent factors capturing shared and specific sources of variation across omics layers.

Master 2 Immunology Research Intern – Immunopathology and Immuno-oncology
CRCL – Centre de Recherche en Cancérologie de Lyon

January 2023 — June 2023
Lyon, France

- Investigated the regulation of NF-κB signaling pathway in response to pro-inflammatory stimuli in human cell lines.
- Performed cell culture, Western blotting, immunofluorescence, and qRT-PCR to assess pathway activation and gene expression.
- Contributed to functional assays exploring the role of NF-κB in cellular stress and apoptosis.
- Analyzed experimental data to characterize molecular mechanisms of inflammation-related gene regulation.

Master 1 Molecular Biology Research Intern – RNA Delivery
CIRI – Centre International de Recherche en Infectiologie

April 2022 — July 2022
Lyon, France

- Developed and characterized a novel RNA delivery system based on the endogenous retroviral protein hPEG10.
- Engineered non-viral particles for mRNA delivery into primary human cells.
- Performed molecular biology techniques including RT-PCR, qPCR, enzymatic digestions, and Western blotting.
- Contributed to the production and analysis of pseudo-viral particles and maintained primary cell cultures.
- Collaborated within a research team and presented results through scientific communication.

EDUCATION

Spring School on AI & Machine Learning in Biology and Health, AI4BioMed, UCBL1

April 7-9, 2025

Bioinformatics Master’s Degree, Claude Bernard Lyon 1 University — (UCBL1)

September 2023 — Present

- Master 2: All courses in Session 1 completed (grades not yet available; no ranking).
- Master 1:
 - Session 1: Grade = 13.213/20, Rank: 12/22.
 - Session 2: Grade = 13.816/20 (no rank available).

(+33) 6 36 85 52 78

Lyon, France

jordan.dutel[AT]lyon.unicancer.fr

Jordan Dutel

Bioinformatics Engineer – Junior Level

Portfolio: github.com/jdutel42

linkedin.com/in/jordan-dutel

Molecular and Cellular Biology Master's Degree, *Claude Bernard Lyon 1 University — (UCBL1)*
(Specialization in Immunology, Immunopathology, & Immunotherapy)

September 2021 — July 2023

- Master 2:
 - Session 1: Grade = 13.225/20, Rank: 10/12.
- Master 1:
 - Session 1: Grade = 13.116/20, Rank: 46/75.

Genetics Bachelor's Degree, *Claude Bernard Lyon 1 University — (UCBL1)*

September 2018 — June 2021

Licence 3:

- Session 1: Grade = 11.763/20, Rank: 39/129.
- Session 2: Grade = 12.509/20, Rank: 6/76.

PACES (Première Année Commune aux Études de Santé), *Jean Monnet University*

September 2016 — June 2018

Level: "Reçus-collés"

Scientific Baccalauréat (Honors: 'Assez Bien'), *Lycée L'Astrée, France*

June 2016

RELEVÉ DE NOTES

Nom de famille : DUTEL
 Nom d'usage :
 Prénoms : JORDAN GILLES YANNICK
 Né(e) le : 28/06/1998
 à : MONTBRISON (042)
 Pays :
 I.N.E. : 1009003865G
 Etablissement : LYCEE L'ASTREE BOEN
 42130 BOEN

M. DUTEL JORDAN GILLES YANNICK
 398 RUE DE CORBES
 42130 MARCILLY-LE-CHATEL

N° matricule : M007938060 N° inscription : 0514454519 Jury : 0079

Centre d'écrit : LYCEE BEAUREGARD MONTBRISON

EPREUVES 1er GROUPE					DISCIPLINES	2è GROUPE			1er GROUPE + 2è GROUPE		
NOTE /20	OBTENUE EN ACA		COEFF.	POINTS		NOTE /20		COEFF.	POINTS 1er GROUPE	POINTS 2è GROUPE	POINTS RETENUS
13	2015 10		2	26	FRANCAIS ECR.						
10	2015 10		2	20	FRANCAIS ORAL						
09			3	27	HIST.GEOL.						
12			7	84	MATHEMATIQUES						
11			8	88	PHYS-CHIMIE						
12			6	72	SC. VIE TERRE						
11			3	33	L.V.E. 1						
					(ANGLAIS)						
17			2	34	LANGUE VIV. 2						
					(ESPAGNOL)						
07			3	21	PHILOSOPHIE						
13			2	26	ED.PHYS.SPORT						
					(EPS APTE)						
16	2015 10			12	TRAV PERS ENC						
17				14	EPR.FACULT.1						
					(ITALIEN)						
TOTAL			38	457		TOTAL			TOTAL		
MOYENNE SUR 20				12.02					MOYENNE SUR 20		

DOCUMENT À CONSERVER.
IL NE SERA DÉLIVRÉ AUCUN DUPLICATA.

B A R E M E	Total 1er groupe	304	380	456	532	608
	Total 2è groupe	304	380			
	Moyenne	8/20	10/20	12/20	14/20	16/20

Le chef de centre ou président du jury
(cachet ou signature)



1ER GROUPE : ADMIS MENTION ASSEZ BIEN

RELEVÉ DE NOTES ET RESULTATS

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Session 1**DUTEL Jordan**

N° Etudiant : 11806616

INE : 1009003865G

Né le : 28 juin 1998

à : MONTBRISON (042)

inscrit en **SEMESTRE 1 L1 PORTAIL PCSI****Notes et résultats**

	Note/Barème	Pts jury	Résultat	Session	Crédits	Rang
Constitution matière	10.45 /20		ADMIS	S1 2018/19	6	218/422
Techniques Maths Base	11 /20		ADMIS	S1 2018/19	6	257/444
CC : Techniques Maths de Base	11 /20			S1 2018/19		
Therm 1 et Transf Thermiq	13.47 /20		ADMIS	S1 2018/19	6	70/426
Physique générale S1 PCSI	14.51 /20		ADMIS	S1 2018/19	6	54/435
TRANSVERSALE 1	16.023 /20		ADMIS	S1 2018/19	6	86/1290
Culture numérique	18.41 /20		ADMIS	S1 2018/19	4	12/1423
Activité sportive 1	11.25 /20		ADMIS	S1 2018/19	2	

Résultat global

Résultat d'admission :	13.091 /20	ADMIS	30	112/407
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Fait à Villeurbanne, le 8 mars 2019
 Le Responsable de la Formation



RELEVÉ DE NOTES ET RESULTATS

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Session 1**DUTEL Jordan**

N° Etudiant : 11806616

INE : 1009003865G

Né le : 28 juin 1998

à : MONTBRISON (042)

inscrit en **SEMESTRE 2 L1 PORTAIL PCSI****Notes et résultats**

	Note/Barème	Pts jury	Résultat	Session	Crédits	Rang
La réaction chimique	12.2 /20		ADMIS	S1 2018/19	6	219/541
Bases de l'Electricité	12.35 /20		ADMIS	S1 2018/19	6	79/435
Maths 2 (Méca, Phys, SPI)	14.05 /20		ADMIS	S1 2018/19	6	109/423
Dynamique S2 PCSI	12.275 /20		ADMIS	S1 2018/19	6	118/419
TRANSVERSALE 2	12.407 /20		ADMIS	S1 2018/19	6	838/1207
Compétences linguistiques 2	8.66 /20		AJOURNE	S1 2018/19		1077/1245
Recherche documentaire	16.62 /20		ADMIS	S1 2018/19	1	131/1345
Projet personnel et professionnel 2	13.75 /20		ADMIS	S1 2018/19	2	
Activité sportive 2	13 /20		ADMIS	S1 2018/19	1	509/1338

Résultat global

Résultat d'admission :	12.656 /20	ADMIS	30	109/386
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Fait à Villeurbanne, le 18 juin 2019
Le Responsable de la Formation



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Session 2**DUTEL Jordan**

N° Etudiant : 11806616

INE : 1009003865G

Né le : 28 juin 1998

à : MONTBRISON (042)

inscrit en **L2 Sciences de la vie****Notes et résultats**

	Note/Barème	Pts jury	Résultat	Session	Crédits	Rang
SEMESTRE 3 L2 SCIENCES DE LA VIE	9.865 /20		AJOURNE	S2 2019/20		134/217
SEMESTRE 4 L2 SCIENCES DE LA VIE	14.382 /20		ADMIS	S1 2019/20	30	80/452
L2 Sciences de la vie	12.124 /20		Admis compensat	S2 2019/20	60	65/243

Résultat global

Résultat d'admission :	12.124/20	ADMIS	65/241
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Fait à Villeurbanne, le 22 mars 2022
Le Responsable de la Formation

RELEVÉ DE NOTES ET RESULTATS

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Session 1**DUTEL Jordan**

N° Etudiant : 11806616

INE : 1009003865G

Né le : 28 juin 1998

à : MONTBRISON (042)

inscrit en **SEMESTRE 3 L2 SCIENCES DE LA VIE****Notes et résultats**

	Note/Barème	Pts jury	Résultat	Session	Crédits	Rang
Enzymologie métabolisme	9.779 /20		AJOURNE	S1 2019/20		138/303
CT : Enzymologie et Métabolisme	8 /20			S1 2019/20		
CC : Enzymologie et Métabolisme	13.93 /20			S1 2019/20		
Intro Bio Cell Immunolog	8.19 /20		AJOURNE	S1 2019/20		176/280
CT : Introduction bio cellule immunologie	7.25 /20			S1 2019/20		
CT : Introduction bio cellule immunologie	9.13 /20			S1 2019/20		
Microbiologie 1	7.72 /20		AJOURNE	S1 2019/20		261/309
CT : Microbiologie 1	8.69 /20			S1 2019/20		
CT : Microbiologie 1	6.75 /20			S1 2019/20		
Biologie du développement	7.818 /20		AJOURNE	S1 2019/20		137/175
CT : Biologie du développement	6.105 /20			S1 2019/20		
CC : Biologie du développement	11 /20			S1 2019/20		
Transversale 3	12.7 /20		ADMIS	S1 2019/20	6	519/1151
Compétences linguistiques 3	13 /20		ADMIS	S1 2019/20	2	672/1206
Projet personnel et professionnel 3	12 /20		ADMIS	S1 2019/20	1	295/1156
Activité sportive 3	12 /20		ADMIS	S1 2019/20	1	919/1314
Ouverture TR3	13.1 /20		ADMIS	S1 2019/20	2	452/1158
Panorama et diversité de la psychologie	13.1 /20		ADMIS	S1 2019/20		

Résultat global

Résultat d'admission :	9.241 /20	AJOURNE	350/477
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Fait à Villeurbanne, le 25 mai 2020
Le Responsable de la Formation



RELEVÉ DE NOTES ET RESULTATS

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Session 2**DUTEL Jordan**

N° Etudiant : 11806616

INE : 1009003865G

Né le : 28 juin 1998

à : MONTBRISON (042)

inscrit en **SEMESTRE 3 L2 SCIENCES DE LA VIE****Notes et résultats**

	Note/Barème	Pts jury	Résultat	Session	Crédits	Rang
Enzymologie métabolisme	9.779 /20		AJOURNE	S2 2019/20		40/164
CT : Enzymologie et Métabolisme	8 /20			S1 2019/20		
CC : Enzymologie et Métabolisme	13.93 /20			S1 2019/20		
Intro Bio Cell Immunolog	8.19 /20		AJOURNE	S2 2019/20		83/162
CT : Introduction bio cellule immunologie	7.25 /20			S1 2019/20		
CT : Introduction bio cellule immunologie	9.13 /20			S1 2019/20		
Microbiologie 1	7.72 /20		AJOURNE	S2 2019/20		83/98
CT : Microbiologie 1	8.69 /20			S1 2019/20		
CT : Microbiologie 1	6.75 /20			S1 2019/20		
Biologie du développement	10.935 /20		ADMIS	S2 2019/20	6	14/86
CT : Biologie du développement	10.9 /20			S2 2019/20		
CC : Biologie du développement	11 /20			S1 2019/20		
Transversale 3	12.7 /20		ADMIS	S1 2019/20	6	521/1155
Compétences linguistiques 3	13 /20		ADMIS	S1 2019/20	2	673/1208
Projet personnel et professionnel 3	12 /20		ADMIS	S1 2019/20	1	295/1156
Activité sportive 3	12 /20		ADMIS	S1 2019/20	1	919/1314
Ouverture TR3	13.1 /20		ADMIS	S1 2019/20	2	452/1158
Panorama et diversité de la psychologie	13.1 /20		ADMIS	S1 2019/20		

Résultat global

Résultat d'admission :	9.865/20	AJOURNE	134/218
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Fait à Villeurbanne, le 23 juillet 2020
Le Responsable de la Formation



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Session 1**DUTEL Jordan**

N° Etudiant : 11806616

INE : 1009003865G

Né le : 28 juin 1998

à : MONTBRISON (042)

inscrit en **SEMESTRE 4 L2 SCIENCES DE LA VIE****Notes et résultats**

	Note/Barème	Pts jury	Résultat	Session	Crédits	Rang
CT Biostatistiques et bioinformatique	16.8 /20		ADMIS	S1 2019/20	6	66/484
Génomés, Santé et Evolution	13.665 /20		ADMIS	S1 2019/20	6	20/60
CT : Génomés santé et évolution GSE	13.4 /20			S1 2019/20		
CC : Génomés santé et évolution GSE	14.062 /20			S1 2019/20		
Génétique 2	12.655 /20		ADMIS	S1 2019/20	6	114/474
CT : Génétique 2	17.56 /20			S1 2019/20		
CT : Génétique 2	7.75 /20			S1 2019/20		
CT Physiologie animale	15.72 /20		ADMIS	S1 2019/20	6	31/343
Transversale 4	13.07 /20		ADMIS	S1 2019/20	6	
Compétences linguistiques 4	12.2 /20		ADMIS	S1 2019/20	2	776/1151
Projet personnel et professionnel 4	Dispense					
Activité sportive 4	Dispense					
Ouvertures TR4	13.94 /20		ADMIS	S1 2019/20	2	483/1155
Ouvertures TR4	13.94 /20		ADMIS	S1 2019/20	2	483/1155

Résultat global

Résultat d'admission :	14.382 /20	ADMIS	30	81/451
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Fait à Villeurbanne le 29 juin 2020
Le Responsable de la Formation



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Session 1**DUTEL Jordan**

N° Etudiant : 11806616

INE : 1009003865G

Né le : 28 juin 1998

à : MONTBRISON (042)

inscrit en **UE L3 SV GENETIQUE ET BIOLOGIE CELLULAIRE****Notes et résultats**

	Note/Barème	Pts jury	Résultat	Session	Crédits	Rang
BLOC 1 AUTOMNE GBC	12.636 /20			S1 2020/21	30	27/132
Biologie cellulaire	13.05 /20		ADMIS	S1 2020/21	6	38/369
CT : Biologie Cellulaire	15.5 /20			S1 2020/21		
CT : Biologie Cellulaire	10.6 /20			S1 2020/21		
Microbiologie 2	13.51 /20		ADMIS	S1 2020/21	6	13/357
CT : Microbiologie 2	13.45 /20			S1 2020/21		
CC : Microbiologie 2	13.75 /20			S1 2020/21		
Outils et bases moléculaires	10.23 /20		ADMIS	S1 2020/21	6	98/375
CT : Outils et bases moléculaires	13.96 /20			S1 2020/21		
CT : Outils et bases moléculaires	6.5 /20			S1 2020/21		
Régulations physiologiques	13.225 /20		ADMIS	S1 2020/21	6	74/360
CT : Régulations Physiologiques	13.333 /20			S1 2020/21		
CC : Régulations Physiologiques	12.5 /20			S1 2020/21		
CT : Régulations Physiologiques	13.25 /20			S1 2020/21		
Transversale 5	13.166 /20		ADMIS	S1 2020/21	6	850/1455
Langues 5	13.666 /20		ADMIS	S1 2020/21	3	710/1464
Activité sportive 5	11 /20		ADMIS	S1 2020/21	1	1332/1459
Ouverture TR5	13.5 /20		ADMIS	S1 2020/21	2	764/1456
Espace et société	13.5 /20			S1 2020/21		29/111
BLOC 2 PRINTEMPS GBC	10.891 /20			S1 2020/21	30	66/135
Biologie cellulaire et pathologies	8.2 /20		AJOURNE	S1 2020/21		79/89
CT : Biologie cellulaire et pathologie	6.25 /20			S1 2020/21		
CC : Biologie cellulaire et pathologie	14.5 /20			S1 2020/21		
CT : Biologie cellulaire et pathologie	10 /20			S1 2020/21		
Dynamique de la cellule	9.161 /20		AJOURNE	S1 2020/21		66/134
CT : Dynamique de la cellule	12.25 /20			S1 2020/21		

Fait à Villeurbanne, le 15 juillet 2021
Le Responsable de la Formation



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Session 1**DUTEL Jordan**

N° Etudiant : 11806616

INE : 1009003865G

Né le : 28 juin 1998

à : MONTBRISON (042)

inscrit en **UE L3 SV GENETIQUE ET BIOLOGIE CELLULAIRE****Notes et résultats**

	Note/Barème	Pts jury	Résultat	Session	Crédits	Rang
CT : Dynamique de la cellule	6.072 /20			S1 2020/21		
Dynamique gènes génomés	11.8 /20		ADMIS	S1 2020/21	6	44/210
TP de biologie cellulaire	14.167 /20		ADMIS	S1 2020/21	6	18/66
CT : Travaux pratiques biologie cellul	14 /20			S1 2020/21		
CC : Travaux pratiques biologie cellul	15 /20			S1 2020/21		
Options S6 GBC	11.125 /20		ADMIS	S1 2020/21		
Intéact entre macromolec	11.125 /20		ADMIS	S1 2020/21	6	36/64
CT : Interactions entre macromolécules	11.5 /20			S1 2020/21		
CT : Interactions entre macromolécules	10.25 /20			S1 2020/21		

Résultat global

Résultat d'admission :	11.763 /20	AJOURNE	39/129
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Fait à Villeurbanne, le 15 juillet 2021
Le Responsable de la Formation



RELEVÉ DE NOTES ET RESULTATS

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Session 2**DUTEL Jordan**

N° Etudiant : 11806616

INE : 1009003865G

Né le : 28 juin 1998

à : MONTBRISON (042)

inscrit en **UE L3 SV GENETIQUE ET BIOLOGIE CELLULAIRE****Notes et résultats**

	Note/Barème	Pts jury	Résultat	Session	Crédits	Rang
BLOC 1 AUTOMNE GBC	12.636 /20			S2 2020/21	30	5/81
Biologie cellulaire	13.05 /20		ADMIS	S1 2020/21	6	38/369
Microbiologie 2	13.51 /20		ADMIS	S1 2020/21	6	13/357
Outils et bases moléculaires	10.23 /20		ADMIS	S1 2020/21	6	98/375
Régulations physiologiques	13.225 /20		ADMIS	S1 2020/21	6	74/360
Transversale 5	13.166 /20		ADMIS	S1 2020/21	6	850/1455
Langues 5	13.666 /20		ADMIS	S1 2020/21	3	710/1464
Activité sportive 5	11 /20		ADMIS	S1 2020/21	1	1332/1459
Ouverture TR5	13.5 /20		ADMIS	S1 2020/21	2	764/1456
Espace et société	13.5 /20			S1 2020/21		29/111
BLOC 2 PRINTEMPS GBC	12.381 /20			S2 2020/21	30	9/76
Biologie cellulaire et pathologies	13.6 /20		ADMIS	S2 2020/21	6	2/32
CT : Biologie cellulaire et pathologie	11.5 /20			S2 2020/21		
CC : Biologie cellulaire et pathologie	14.5 /20			S1 2020/21		
CT : Biologie cellulaire et pathologie	17.5 /20			S2 2020/21		
Dynamique de la cellule	11.215 /20		ADMIS	S2 2020/21	6	2/66
CT : Dynamique de la cellule	12.25 /20			S1 2020/21		
CT : Dynamique de la cellule	10.18 /20			S2 2020/21		
Dynamique gènes génomes	11.8 /20		ADMIS	S1 2020/21	6	44/210
TP de biologie cellulaire	14.167 /20		ADMIS	S1 2020/21	6	18/66
Options S6 GBC	11.125 /20		ADMIS	S1 2020/21		
Intéact entre macromolec	11.125 /20		ADMIS	S1 2020/21	6	36/64

Résultat global

Résultat d'admission :	12.509/20	ADMIS	60	6/76
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Fait à Villeurbanne, le 15 juillet 2021
Le Responsable de la Formation



RELEVÉ DE NOTES ET RESULTATS

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Session 1

DUTEL Jordan
398 Rue de Corbes
42130 MARCILLY LE CHATEL

DUTEL Jordan

N° Etudiant : 11806616

INE : 1009003865G

Né le : 28 juin 1998

à : MONTBRISON (042)

inscrit en **M1 Bio moléculaire et cel****Notes et résultats**

	Note/Barème	Pts jury	Résultat	Session	Crédits	Rang
Sem1 M1 Bio. mol. & cel.	13.448 /20		ADMIS	S1 2021/22	30	23/76
Sem2 M1 Bio. mol. & cel.	12.784 /20		ADMIS	S1 2021/22	30	61/75

Résultat global

Résultat d'admission :	13.116 /20		ADMIS		60	46/75
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Fait à Villeurbanne, le 20 juin 2022
Le Responsable de la Formation

RELEVÉ DE NOTES ET RESULTATS

Page : 1 / 1

Session 1

DUTEL Jordan
398 Rue de Corbes
42130 Marcilly-Le-Châtel
42130 MARCILLY LE CHATEL

DUTEL Jordan

N° Etudiant : 11806616

INE : 1009003865G

Né le : 28 juin 1998

à : MONTBRISON (042)

inscrit en **M2 Immunol immunopath****Notes et résultats**

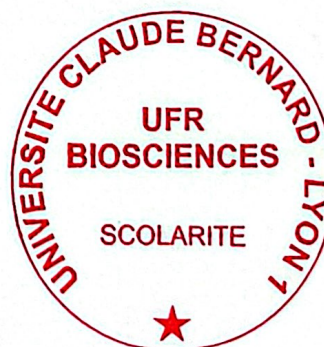
	Note/Barème	Pts jury	Résultat	Session	Crédits	Rang
M2 Immunol immunopath	13.333 /20		ADMIS	S1 2022/23		12/12
SEMESTRE 1 - M1 BMC					(1)	
SEMESTRE 2 - M1 BMC					(2)	
SEMESTRE 3 - M2 IMMUNO	12.916 /20		ADMIS	S1 2022/23	30	
SEMESTRE 4 - M2 IMMUNO	13.75 /20		ADMIS	S1 2022/23	30	

Résultat global

Résultat d'admission :	13.225 /20	ADMIS	120	10/12
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Détail des résultats utilisés pour les éléments du relevé de notes :

	Note/Barème	Pts jury	Résultat	Session	Crédits	Rang
(1) Semestre 1 M1 Biologie moléculaire et cellulaire	13.448 / 20		ADMIS	S1 2021/22	30	
(2) Semestre 2 M1 Biologie moléculaire et cellulaire	12.784 / 20		ADMIS	S1 2021/22	30	



Fait à Villeurbanne, le 28 juin 2023
Le Responsable de la Formation

RELEVÉ DE NOTES ET RESULTATS

Session 1

DUTEL Jordan

N° Etudiant : 11806616

INE : 1009003865G

Né le : 28 juin 1998

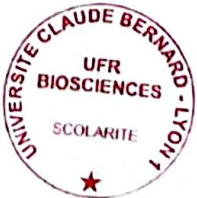
à : MONTBRISON (042)

inscrit en M1 Bio-informatique

Notes et résultats

	Note/Barème	Pts jury	Résultat	Session	Crédits
AMDB I	12.759 /20		ADMIS	S1 2023/24	27
Bioinfo structurale	10.2 /20		ADMIS	S1 2023/24	6
CT	9 /20			S1 2023/24	
CC	12 /20			S1 2023/24	
Bases bioinfo mol	13.29 /20		ADMIS	S1 2023/24	6
CT	15.75 /20			S1 2023/24	
CC	10.875 /20			S1 2023/24	
CT	13.17 /20			S1 2023/24	
MADP	15.833 /20		ADMIS	S1 2023/24	3
CT	15.833 /20			S1 2023/24	
MADG	12.504 /20		ADMIS	S1 2023/24	6
CT	12.723 /20			S1 2023/24	
CC	11.63 /20			S1 2023/24	
MADT	13.506 /20		ADMIS	S1 2023/24	6
CT	12.757 /20			S1 2023/24	
CC	16.5 /20			S1 2023/24	
GVDB I	10.227 /20		AJOURNE	S1 2023/24	
BDBIO	12.04 /20		ADMIS	S1 2023/24	3
PAPB	9.32 /20		AJOURNE	S1 2023/24	
CMPADB I	14.991 /20		ADMIS	S1 2023/24	18
MPB	13.694 /20		ADMIS	S1 2023/24	3
CT	13.694 /20			S1 2023/24	
Projet bioinfo. 1	12.416 /20		ADMIS	S1 2023/24	3
CC	12.416 /20			S1 2023/24	
Projet bioInfo. 2	15.833 /20		ADMIS	S1 2023/24	3
CC	15.833 /20			S1 2023/24	
Stage bioinfo.1	16 /20		ADMIS	S1 2023/24	9
CT	16 /20			S1 2023/24	
CIPR I	14.405 /20		ADMIS	S1 2023/24	6
Gestion de projet	16 /20		ADMIS	S1 2023/24	3
CCI	16 /20			S1 2023/24	

Fait à Villeurbanne, le 15 juillet 2024
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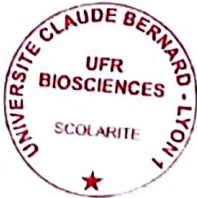
RELEVÉ DE NOTES ET RESULTATS

Session 1

DUTEL Jordan
N° Etudiant : 11806616 INE : 1009003865G
Né le : 28 juin 1998 à : MONTBRISON (042)
inscrit en **M1 Bio-informatique**

Notes et résultats					
	Note/Barème	Pts jury	Résultat	Session	Crédits
Anglais niveau 1					(1)
Résultat global					
Résultat d'admission :	13.213 /20	AJOURNE			12/22
Détail des résultats utilisés pour les éléments du relevé de notes :					
	Note/Barème	Pts jury	Résultat	Session	Crédits
(1) Anglais pour la communication professionnelle niveau 1	12.81 / 20		ADMIS	S1 2021/22	3

Fait à Villeurbanne, le 15 juillet 2024
Le Responsable de la Formation



RELEVÉ DE NOTES ET RESULTATS

Session 2

DUTEL Jordan

N° Etudiant : 11806616

INE : 1009003865G

Né le : 28 juin 1998

à : MONTBRISON (042)

inscrit en M1 Bio-informatique

Notes et résultats

	Note/Barème	Pts jury	Résultat	Session	Crédits
AMDB I	12.759 /20		ADMIS	S1 2023/24	27
Bioinfo structurale	10.2 /20		ADMIS	S1 2023/24	6
Bases bioinfo mol	13.29 /20		ADMIS	S1 2023/24	6
MADP	15.833 /20		ADMIS	S1 2023/24	3
MADG	12.504 /20		ADMIS	S1 2023/24	6
MADT	13.506 /20		ADMIS	S1 2023/24	6
GVDB I	14.247 /20		ADMIS	S2 2023/24	9
BDBIO	12.04 /20		ADMIS	S1 2023/24	3
PAPB	15.35 /20		ADMIS	S2 2023/24	6
CMPADB I	14.991 /20		ADMIS	S1 2023/24	18
MPB	13.694 /20		ADMIS	S1 2023/24	3
Projet bioinfo. 1	12.416 /20		ADMIS	S1 2023/24	3
Projet bioinfo. 2	15.833 /20		ADMIS	S1 2023/24	3
Stage bioinfo.1	16 /20		ADMIS	S1 2023/24	9
CIPR I	14.405 /20		ADMIS	S1 2023/24	6
Gestion de projet	16 /20		ADMIS	S1 2023/24	3
Anglais niveau 1					(1)

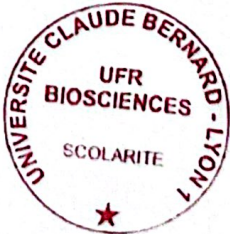
Résultat global

Résultat d'admission :	13.816/20	ADMIS	60
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Détail des résultats utilisés pour les éléments du relevé de notes :

	Note/Barème	Pts jury	Résultat	Session	Crédits
(1) Anglais pour la communication professionnelle niveau 1	12.81 / 20		ADMIS	S1 2021/22	3

Fait à Villeurbanne, le 15 juillet 2024
Le Responsable de la Formation





Sabine Peres, Fabien Duchateau
Université Claude Bernard Lyon 1
04 72 44 58 25
{prénom.nom}@univ-lyon1.fr

Lyon, 15/04/2025

À qui de droit

Obj. : Attestation de notes pour **Jordan Dutel**

Madame, Monsieur,

Nous sommes enseignants-chercheurs à l'Université Claude Bernard Lyon 1 et responsables du [Master 2 Bioinformatique](#).

Par la présente, nous attestons des notes de Jordan Dutel au Master Bioinformatique de Lyon, extraites du logiciel de notes de l'UCBL (voir page suivante).

Toutes les notes ne sont pas encore disponibles, et ces notes n'ont pas encore été validées par le jury. Le procès-verbal définitif ne sera établi que dans quelques mois.

Si vous avez des questions, n'hésitez pas à nous contacter (de préférence par email).

Bien cordialement,

Sabine Peres, Fabien Duchateau
Enseignants-chercheurs
Responsables du M2 Bioinformatique



▼ 2MBIH1BA

▼ UE-BIO1417M Génomique En Écologie Et Évolution 1 Marie FABLET

Parcours BioInfo	Marqueur NADH6	Nb aa total 186	Nb aa analyse 33
Br 4 65	Br 5 74	Br 6 V+S+R 83	Br 7 89
Fungi + microsporidia 1	Alveolate 1	Ciliates 1	Alveolate+ 1
Moyenne 12.25/20	CT Landscape 14.00/20	CT phylo 10.50/20	

▼ 2MBIH1BB

▼ UE-BIO2530M Visualisation De Données Biologiques Guillaume LAUNAY

total 16.00/20

▼ UE-INF2409M Programmation Web Pour La Bioinformatique Guillaume LAUNAY

Ergonomie matin PRST	Ergonomie après-midi PRST	exam 15.00/20
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▼ 2MBIH1BC

▼ UE-BIO2224M Statistiques Bayésiennes Et Applications Laurent GUEGUEN

▼ UE-BIO2461M Stage Entreprise / Laboratoire 2 Celine BROCHIER ARMANET

▼ UE-BIO2529M Algorithmique Avancée Pour La Bio-Informatique Arnaud MARY

▼ UE-INF2407M Projet En Bioinformatique 3 Laurent GUEGUEN

Note 14.666/20	Soutenance 12.00/20	CoC 16.00/20	Delivrable 16.00/20
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▼ UE-INF2417M Techniques D'Apprentissage Automatique Khalid BENABDESLEM, Haytham ELGHAZEL

TP1 16/12/2024 PRST	Présence TP SemiFS PRST		
P3 CCI 14.400/20	TP Khalid 15.50/20	TP ELGHAZEL 14.50/20	Examen TP 13.50/20

▼ UE-INF2481M Analyse De Données Fabien DE MARCHI, Remy CAZABET

Dash fichiers Vérifier Dash_app.zip 4136.33KB Remplacer ce fichier	Dash URL https://github.com/jdutel42/	projet 14.00/20
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▼ 2MBIH1BD

▼ UE-BIO2678M Enjeux Éthiques-Juridiques-Sociaux-Écologiques En Bioinfo Vincent LACROIX

▼ UE-BIOLG02M Anglais Pour La Communication Professionnelle Niveau 2 Nathalie DOURLLOT

essay M2 bioinfo 14.50/20	listening test M2 bioinfo 14.00/20
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Laboratory address:
CRCL, Equipe Saintigny
Cheney B, 2^e étage
28 rue Laënnec
69373 Lyon Cedex 08
FRANCE

Dr. Pierre Martinez
Bioinformatics of Cancer Evolutionary Dynamics

Nationality: French
Birth date: July 6th 1984

<https://pierremartinez.github.io>
pierre.martinez [AT] lyon.unicancer.fr
+33 4 69 83 60 82

EDUCATION

- 2023 **HDR** (*Habilitation à Diriger des Recherches*), Université Claude Bernard Lyon 1.
- 2008-2011 **PhD** under the supervision of Prof. Franca Fraternali (Randall Division of Cell and Molecular Biophysics, King's College London, England).
"Computational Analysis of Transcription Factors Cooperation and Protein-Protein Interactions Structural Features". *CASE Award (EPSRC)* in collaboration with the Nestlé Research Centre (Dr James Holzwarth, Lausanne, Switzerland).
- 2007 **Master in Bioinformatics**
Sardegna Ricerche, Pula, Italy.
- 2001-2006 **Master in Information Technology**
EPITECH, Paris, France.

WORK EXPERIENCE

- 2018- **Chargé de Recherche (Permanent PI position)**
Inserm / Cancer Research Centre of Lyon, France.
Quantifying, modeling, predicting and controlling cancer evolution and diversity.
- 2016-2018 **Assistant Scientist**
Cancer Research Centre of Lyon / Centre Léon Bérard, Lyon, France.
Analysis of clonal evolution of tumours.
- 2014-2016 **Post-doctoral Research Assistant**
Evolution and Cancer, Dr Trevor Graham's lab, Barts Cancer Institute, Queen Mary University, London, UK.
Analysis of evolutionary processes and dynamics in tumours and pre-malignant conditions.
- 2011-2013 **Post-doctoral Research Assistant**
Translational Cancer Therapeutics, Prof. Charles Swanton's lab, Cancer Research UK, London, UK.
Analysis of intra-tumour heterogeneity and chromosomal instability.

SELECTED PUBLICATIONS in the last 5 years

Muller L, Fauvet F, Chassot C, Angileri F, **Coutant A**, [...], **Martinez P**. EMT-driven plasticity prospectively increases cell-cell variability to promote therapeutic adaptation in breast cancer. *Cancer Cell International*, 2025.

Coutant A, Cockenpot V, Muller L, [...], Ouzounova M*, **Martinez P***. Spatial transcriptomics reveal pitfalls and opportunities for the detection of rare high-plasticity breast cancer subtypes. *Laboratory Investigation*, 2023.

Monteiro L, Da Silva L, Lipinski B, [...], **Martinez P**. Assessing Cell Activities rather than Identities to Interpret Intra-Tumor Phenotypic Diversity and Its Dynamics. *iScience*, 2020.

Tokutomi N, Moyret-Lalle C, Puisieux A, Sugano S, **Martinez P**. Quantifying the fitness of tissue-specific evolutionary trajectories by harnessing cancer's repeatability at the genetic level. *Evolutionary Applications*, 2019.

Beddok A, Velleuer, E, [...], Martinez P, [...], Thariat J. Strategies for early detection and detailed characterization of oral lesions and head and neck squamous cell carcinoma in Fanconi anemia patients. *Cancer Letters*, 2025.

Gatenbee C, Baker AM, Schenck R, [...], Martinez P, [...], Graham TA, Anderson, AR. Immunosuppressive niche engineering at the onset of human colorectal cancer. *Nature Communications*, 2022.

Truchard E, Bertolus C, Martinez P, Thomas E, Saintigny P, Foy JP. Identification of a Gene-Expression-Based Surrogate of Genomic Instability during Oral Carcinogenesis. *Cancers*, 2022.

Bouaoud J, [...], Martinez P, [...], Saintigny P. Unmet Needs and Perspectives in Oral Cancer Prevention. *Cancers*, 2022.

Pommier RM, Sanlaville A, Tonon L, Kielbassa J, Thomas E, Ferrari A, Sertier AS, Hollande F, Martinez P, Tissier A, Morel AP, Ouzounova M & Alain Puisieux. Comprehensive characterization of claudin-low breast tumors reflects the impact of the cell-of-origin on cancer evolution. *Nature Communications*, 2020.

TEACHING

Lectures on Evolutionary trajectories in cancer, Master 2 de Cancérologie, Université Paris Saclay: 2h per year (2020-).

Demonstrations for **structural bioinformatics courses**, undergraduates King's College London: 6 hours (2009).

Lectures on cancer evolution, year 1 medical students Queen Mary University of London: 1 hour per year (2014-2016).

Problem-based learning, Human Science and Public Health module, year 2 medical students, Queen Mary university of London: 14 hours (2016).

Supervision of **8 Master students** in Bioinformatics and **2 undergraduate students** (laboratory technicians).

Main supervisor of **1 PhD student** (started November 2020).

SCIENTIFIC RESPONSIBILITIES

Member of the "**Technological Innovations for Health**" commission for the regional *Cancéropôle Lyon Auvergne Rhône Alpes* Scientific Committee, 2022.

Member of the organising committee for the **Dynamics and statistics of cancer evolution workshop**, Marseille, France, June 2022.

Member of the organising committee for the "**Mathematical Models of Cancer Evolution and Ecology**" symposium during the Mathematical Models of Evolution and Ecology conference, Lyon, France, July 2019.

Member of the organising committee for the **3rd International Symposium of the Cancer Research Centre of Lyon**, Lyon, France, September 2017.

SCIENTIFIC COMMUNICATIONS

Conférence de la Société Française de Médecine Préventive et Personnalisée, 2025, Paris : Tabac, cancer, maladie de Fanconi et évolution somatique de la muqueuse orale. *Invited speaker*

Oral Pre-Malignant Disease Day, 2025, Zagreb, Croatia: Oral mucosa evolutionary dynamics in health and disease. *Invited speaker*

Single Cell Symposium, 2024 Ankara, Turkey: Single-cell analyses of intra-tumour phenotypic heterogeneity and cellular plasticity. *Invited speaker*

RECOMB Satellite Workshop on Computational Cancer Biology, 2023 Istanbul, Turkey: Spatial transcriptomics reveal pitfalls and opportunities for the detection of rare high-plasticity breast cancer subtypes. *Contributed talk*.

Hétérogénéité tumorale : du séquençage de l'ADN aux applications cliniques, 2023, Société Française du Cancer, Paris, France: "Quantification de l'hétérogénéité intra-tumorale". *Invited talk*.

CRCL International symposium, 2019, Lyon, France: "Harnessing cancer's repeatability at the genetic level". *Contributed talk*.

IARI symposium, March 2019, Tokyo, Japan: "Harnessing cancer's repeatability at the genetic level". *Invited speaker*.

The **Cancer Mosaic, Traits, Strategies and Adaptations Workshop**, August 2018, Montpellier, France: "Mapping pre-neoplastic development using contingency-based fitness". *Invited speaker*.

International Society for Evolution, Ecology and Cancer Conference, Tempe AZ, USA, 2017: "Evolution of Barrett's Esophagus through space and time at whole-biopsy and single-crypt levels". *Contributed talk*.

European Conference on Mathematical and Theoretical Biology, July 2016, Nottingham, UK: "Predetermined malignant potential in Barrett's Oesophagus". *Invited speaker (satellite meeting)*.

Nottingham Pathology 2016, June 2016, Nottingham, UK: "Predetermined malignant potential in Barrett's Oesophagus". *Contributed talk*.

Research Progress in Barrett's Oesophagus 9th National Meeting, April 2016, London, UK: "Measuring and modelling evolutionary clonal dynamics in Barrett's Oesophagus to predict cancer risk". *Contributed talk*.

From gene to phenotype: recent advances in biology and medicine workshop, March 2016, Warsaw, Poland. "Pan-cancer analysis of intratumoral heterogeneity". *Invited Speaker*.

Third International Biannual Evolution and Cancer Conference, December 2015, San Francisco, USA: "Predetermined malignant potential in Barrett's Oesophagus". *Contributed talk*.

Cancer Research Centre of Toulouse seminar, November 2015, Toulouse, France: "Quantifying cancer evolution and intratumour heterogeneity". *Invited speaker*.

National Cancer Research Institute (NCRI) Cancer Conference, November 2015, Liverpool, UK: "Longitudinal single cell clonal analysis reveals evolutionary stasis and predetermined malignant potential in non-dysplastic Barrett's Oesophagus". *Contributed talk*.

Evolution and Cancer Conference, Mars 2015, Montpellier, France: "Tracking clonal evolution over time in Barrett's Oesophagus". *Invited speaker*.

Research Progress in Barrett's Oesophagus 8th National Meeting, April 2014, London, UK: "Measuring the pattern of clonal evolution in Barrett's Oesophagus using single-cell FISH data in a large longitudinally-followed cohort". *Contributed talk*.

FUNDING

Projects

2025: Ruban Rose MAESTRO. Team 4 co-leader 75K€ (225K€ total).

2024: Inter-SIRIC collaboration. Team 3 co-leader 50K€ (150K€ total).

2023: Ligue Contre le Cancer « Multi-team project » Project and Team 1 manager €25K (68K total).

2023: Agence Nationale de la Recherche « PRCI ». Project and Team 1 manager €172K (322K total).

2022: Institut National du Cancer « PL-BIO ». Project and Team 1 manager, €224k (412k total).

2021: Institut National du Cancer « PREVBIO ». Team 1 co-manager, €367k (415k total).

2020: Cancéropôle Lyon Auvergne Rhône Alpes « Oncostarter numérique ». Project manager, €40k.

2019: Plan Cancer Single-Cell. Project manager, €360k.

2019: Plan Cancer Maths, Informatics & Cancer. Team 2 manager, €370k (570k total).

Other

2022: Ligue Nationale Contre le Cancer « Subvention colloque » (2,375€).

2019: Cancéropôle Lyon Auvergne Rhône Alpes « Soutien aux manifestations scientifiques » (1,000€).

OTHER

2023 Scientific coordinator of the DEMO clinical protocol (NCT06094829)

2024 Scientific coordinator of the TACO clinical protocol (NCT06425146)

4) Project

Evolutionary determinants and trajectories of metaplastic breast carcinomas

Introduction

Triple-negative breast cancers (TNBCs) face a lack of therapeutic options and high mortality rates¹. **Metaplastic breast carcinomas (MpBCs) are a rare subtype of TNBCs, characterized by high cellular plasticity** and the presence of at least one transdifferentiated tumor compartment^{2,3} (**Figure 1**). These compartments can be of various non-epithelial types, whose presence is determined exclusively through histological analysis. In addition, the mechanisms underlying their emergence and phenotypic switch remain unexplained. Cellular plasticity, which allows for these dynamic phenotypic changes, may also contribute to therapeutic resistance through non-genetic mechanisms^{4,5}. Due to their rarity and complexity, the biology of MpBCs remains poorly understood, and patients are therefore still inadequately managed.

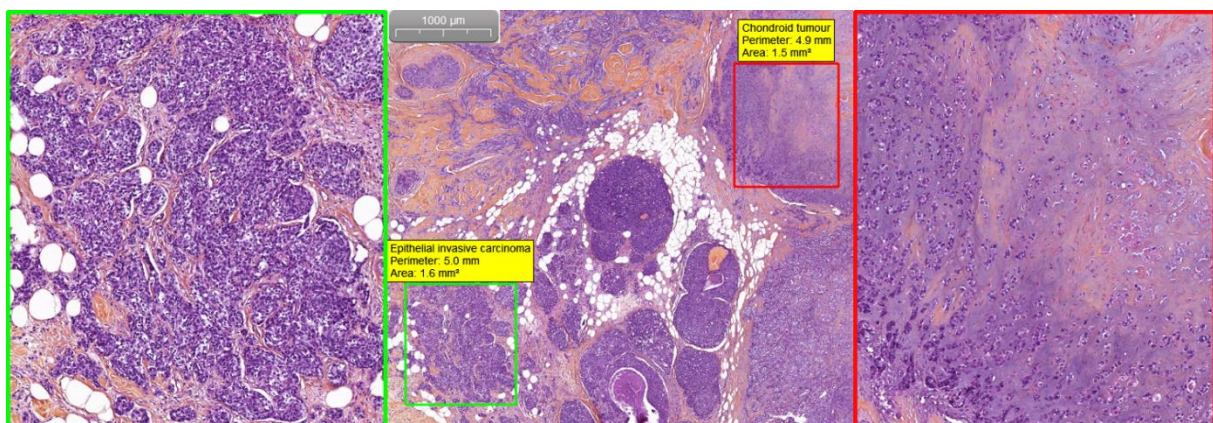


Figure 1 – Example of a mixed MpBC sample.

Within the same tumor, an area of invasive epithelial breast carcinoma is visible on the left (magnified in the green box), and an area of chondroid transdifferentiation is visible on the right (magnified in the red box).

The objectives of this project are to **better understand the emergence of transdifferentiated compartments in MpBCs and to provide molecular diagnostic options for these tumors**. I will focus on mixed-histology MpBCs, in which phenotypically distinct compartment pairs can be studied, using high-precision multi-omics technologies: spatial transcriptomics and single-cell RNA sequencing (scRNA-seq); exome and methylome sequencing after microdissection. This approach will help uncover the origin and genetic/non-genetic mechanisms underlying **four different types of transdifferentiation: spindle cell, squamous, chondroid, and osteoid**. Fifteen mixed MpBC samples have already been analyzed using Visium spatial transcriptomics, and 15 others are currently being processed. The availability of orthogonal scRNA-seq data will enable more precise identification of deregulated genes and signaling pathways in transdifferentiated MpBC cells—both in comparison to classical epithelial cancer cells and to normal mesenchymal cells they resemble histologically. This will lay the groundwork for identifying new molecular diagnostic and classification solutions, as well as therapeutic opportunities that will be validated by the team's collaborators.

To achieve its goals, the project is structured around three specific aims:

- 1) **Creation** of a single-cell and spatial resolution transcriptomic atlas dedicated to MpBC.
- 2) **Characterization** of the genetic and non-genetic determinants of transdifferentiation in MpBC.
- 3) **Identification** of subtype-specific expression markers and signaling pathways.

Creation of a Single-Cell and Spatial Transcriptomic Atlas for MpBC

My project will enable a comprehensive characterization of the cellular and molecular features underlying MpBC heterogeneity. Despite the existence of multiple transcriptomic datasets on inter- and intra-tumoral heterogeneity in breast cancers, none are specific to MpBC. As shown by our preliminary data, this limits the ability of existing datasets to accurately deconvolute spatial transcriptomics data. Due to a lack of appropriate reference data, algorithms incorrectly attribute RNA profiles from Visium “spots” annotated as spindle to fibroblasts (**Figure 2A-C**). Despite the existence of expression signatures to detect the epithelial-to-mesenchymal transition (EMT), we still lack dedicated data on transdifferentiated cells in these breast cancers. The goal is therefore to **create an atlas, i.e. a precise and appropriate reference dataset—for decoding tumor cell heterogeneity in MpBC**, using single-cell RNA sequencing (scRNA-seq) combined with spatial transcriptomics.

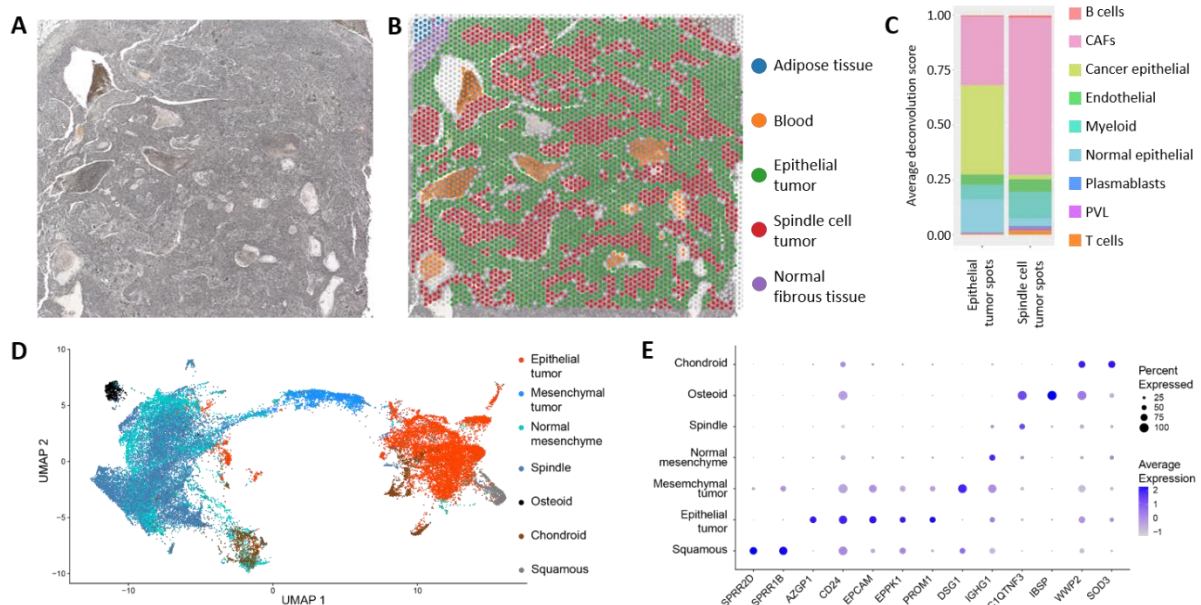


Figure 2 – Preliminary spatial transcriptomic data.

A) Pre-Visium microscopic image of a mixed MpBC sample with spindle cells.

B) Visium spots (~10 cells) annotated.

C) Deconvolution scores of the different cell types contributing to the global transcriptomic signal of epithelial and spindle tumor spots.

D) UMAP projection of annotated spots from 15 mixed MpBC samples. Although subtype-specific clusters can be observed, the separation is imperfect due to Visium spots potentially containing mixtures of cell types.

E) Putative subtype-specific expression markers identified from preliminary analyses of annotated Visium spots. Higher-resolution single-cell data may yield even more relevant biomarkers.

As in previous work by the team⁶ and in my preliminary analyses, Visium slides will be annotated individually using a hybrid approach, combining K-means clustering with manual annotation by a trained pathologist. The **snPATHO-seq** protocol⁷, jointly developed by the single-cell platform at the Centre Léon Bérard (CLB), will be implemented to obtain single-nuclei RNA sequencing (snRNA-seq) data—a scRNA-seq variant better suited to FFPE samples used here. This method allows both **cell type annotation and copy number alteration (CNA) profiling**, with precision similar to cell-based methods. We will use it to create a single-cell expression dataset covering all rare MpBC cell types. For each of

the 4 types of transdifferentiation studied, I will select the 5 best spatially profiled samples (n=20), based on Visium quality control metrics and the clarity of separation between compartments.

A limitation of snRNA-seq is the lack of cross-referenced datasets for accurate annotation, especially for the rare subtypes studied here. However, spatial transcriptomic data and pathologist annotations will help overcome this. Marker expression profiles from snRNA-seq clusters will be cross-examined against dimensionally-reduced projections of our spatial transcriptomics data (**Figure 2D-E**), to confirm their match with specific transdifferentiation states.

Characterization of the genetic and non-genetic determinants of transdifferentiation in MpBC

To better understand the evolutionary trajectories of the 20 selected mixed MpBC samples, the team and its collaborators (Dr. M. Ouzounova, CRCL) are microdissecting the samples using a laser capture microscope (Zeiss PALM MicroBeam, CLB research pathology platform). This will allow **precise identification of genetic features in both compartments of each sample (Figure 3)**.

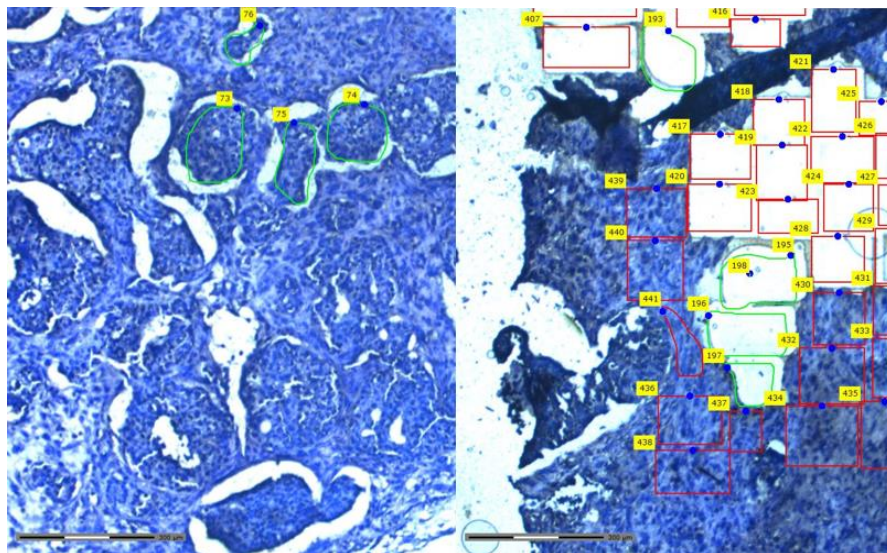


Figure 3 – Laser capture microdissection.

Selection of epithelial (green) and mesenchymal (red) areas in a case of mixed MpBC. Cresyl violet staining. Microdissected fragments from the two compartments are collected into separate tubes, and 8–10 serial sections are microdissected for each sample. Replicates from the same compartment are then pooled per sample for compartment-specific DNA extraction.

Somatic genetic alterations—point mutations, CNAs, and indels—will be identified through whole-exome sequencing (Twist Biosciences 2.0 kit) at 200x coverage. The Mutect2 software⁸ will be used for point mutation and indel analyses. In the absence of a normal control (blood or healthy tissue), paired compartments will serve as each other's reference to identify compartment-specific mutations. A mutation will be considered specific if present (allele frequency >15%) in one compartment and entirely absent in the other. Mutations listed as “drivers” in the COSMIC database and with >15% allele frequency in both compartments will be considered clonal (present in the original cancer cell). PureCN⁹ will be used to identify compartment-specific CNAs, using a pool of normal samples already sequenced by the team as reference.

To explore potential non-genetic determinants of transdifferentiation, deconvolution analyses will be performed using RCTD¹⁰, combining the existing breast cancer atlas data¹¹ and our dedicated MpBC atlas. **Differences in tumor microenvironment (TME) composition will be evaluated** by comparing spot compositions for each (trans)differentiation type, using ANOVA. To investigate whether external factors drive transdifferentiation, we will specifically analyze the composition of inner and outer borders of spatially segregated clusters (>25 spots) associated with transdifferentiated cells. Finally, using ligand-receptor databases^{12–14} and the SpatialDM¹⁵ tool, we will focus on **identifying ligand-receptor pairs likely to indicate interactions** between non-cancerous cells and transdifferentiated cancer cells.

Identification of subtype-specific expression markers and signaling pathways

Despite the existence of transcriptomic signatures like the claudin-low subtype^{16,17}, associated with MpBC¹⁸, the team's previous work has shown that **molecular approaches for MpBC identification remain inadequate**. This is especially due to similarities between mesenchymal stroma and transdifferentiated cells⁶. Differential expression analyses will therefore be conducted to identify specific biomarkers for each compartment type, based on snRNA-seq data (rather than the less accurate Visium data). Normalization and integration will be performed using Seurat. Two different methods will be implemented and compared to identify compartment-specific markers: 1) Seurat's FindAllMarkers function, and 2) MAST¹⁹, which uses a Hurdle model to correct for fixed and random effects, thereby efficiently addressing batch and sample biases. This approach ensures that pathological annotations from our Visium data provide reliable orthogonal information for single-cell classification, which in turn enables **more precise biomarker identification**.

We will also seek to **identify key regulators** potentially driving differential expression in each MpBC subtype. Using the SCENIC tool²⁰, we will identify gene regulatory networks specific to transdifferentiation subtypes and the transcription factors responsible for the observed expression profiles. This should narrow down the list of candidate genes for in-depth study and overcome the challenge of identifying relevant but lowly expressed genes. Finally, we will combine differential expression analyses performed across similar subtypes and within individual mixed-histology tumors. This will ensure that the selected genes are representative of most or all samples within each subtype, rather than being statistically significant but only relevant to a subset of samples.

Impact, feasibility, and perspectives

MpBC is a rare subtype of breast cancer that is still poorly managed. By addressing this issue, my project will benefit both researchers and healthcare professionals working on MpBC, as well as the affected patients. I will create and share a transdifferentiation atlas of MpBC at single-cell resolution with paired spatial resolution. This will **fill a significant knowledge gap** and serve as a valuable resource for future research.

My PhD project is part of a **national initiative** involving the Cancer Research Center of Lyon and the Centre Léon Bérard (Lyon), the Institut Curie (Paris), the Bergonié Institute (Bordeaux), and the Cancer Research and Immunology Center (Nantes). This initiative aims to improve the management of patients with metastatic breast cancer and to develop PDX and organoid models for studying transdifferentiation in aggressive breast cancer forms. This national effort is **supported by Ruban Rose, Ligue Régionale Contre le Cancer, and InterSIRIC funding**, which has enabled the administrative and logistical setup (legal agreements and material transfer between institutions), as well as the production of the multi-omics data I will analyze.

Finally, this national collaboration will enable the clinical and functional validation of the biomarkers I will identify. In collaboration with Institut Curie (Dr. V. Cockenpot), the specificity of biomarkers for each transdifferentiation type will be **validated by immunohistochemistry on an independent cohort**. This will assess their relevance for diagnostic and classification purposes in molecular analyses. Lastly, thanks to the InterSIRIC partnership, the functional impact of these markers will be studied through **genetic and pharmacological perturbations in two patient-derived MpBC models**: mouse xenografts (Dr. E. Marangoni, Institut Curie) and organoids (Dr. V. Guen, Cancer Research and Immunology Center of Nantes).

Références

1. Sung, H. *et al.* Global Cancer Statistics 2020: GLOBOCAN Estimates of Incidence and Mortality Worldwide for 36 Cancers in 185 Countries. *CA. Cancer J. Clin.* **71**, (2021).
2. Reddy, T. P. *et al.* A comprehensive overview of metaplastic breast cancer: clinical features and molecular aberrations. *Breast Cancer Res* **22**, 1–11 (2020).
3. Weigelt, B. *et al.* Metastatic breast carcinomas display genomic and transcriptomic heterogeneity. *Mod. Pathol.* **28**, 340–351 (2014).
4. Shaffer, S. M. *et al.* Rare cell variability and drug-induced reprogramming as a mode of cancer drug resistance. *Nature* **546**, 431–435 (2017).
5. Muller, L. *et al.* EMT-driven plasticity prospectively increases cell–cell variability to promote therapeutic adaptation in breast cancer. *Cancer Cell Int.* **25**, 1–15 (2025).
6. Coutant, A. *et al.* Spatial transcriptomics reveal pitfalls and opportunities for the detection of rare high-plasticity breast cancer subtypes. (2023) doi:<https://doi.org/10.1016/j.labinv.2023.100258>.
7. Wang, T. *et al.* snPATHO-seq, a versatile FFPE single-nucleus RNA sequencing method to unlock pathology archives. *Commun. Biol.* **7**, 1–12 (2024).
8. Cibulskis, K. *et al.* Sensitive detection of somatic point mutations in impure and heterogeneous cancer samples. *Nat. Biotechnol.* **31**, 213–219 (2013).
9. Riester, M. *et al.* PureCN: Copy number calling and SNV classification using targeted short read sequencing. *Source Code Biol. Med.* **11**, 1–13 (2016).
10. Cable, D. M. *et al.* Robust decomposition of cell type mixtures in spatial transcriptomics. *Nat. Biotechnol.* **40**, 517 (2022).
11. Wu, S. Z. *et al.* A single-cell and spatially resolved atlas of human breast cancers. *Nat. Genet.* **53**, 1334–1347 (2021).
12. Efremova, M., Vento-Tormo, M., Teichmann, S. A. & Vento-Tormo, R. CellPhoneDB: inferring cell–cell communication from combined expression of multi-subunit ligand–receptor complexes. *Nat. Protoc.* **15**, 1484–1506 (2020).
13. Noël, F. *et al.* Dissection of intercellular communication using the transcriptome-based framework ICELLNET. *Nat. Commun.* **12**, 1–16 (2021).
14. Jin, S. *et al.* Inference and analysis of cell–cell communication using CellChat. *Nat. Commun.* **12**, 1–20 (2021).
15. Li, Z., Wang, T., Liu, P. & Huang, Y. SpatialDM for rapid identification of spatially co-expressed ligand–receptor and revealing cell–cell communication patterns. *Nat. Commun.* **14**, 1–12 (2023).
16. Hennessy, B. T. *et al.* Characterization of a naturally occurring breast cancer subset enriched in epithelial-to-mesenchymal transition and stem cell characteristics. *Cancer Res.* **69**, 4116–4124 (2009).
17. Prat, A. *et al.* Phenotypic and molecular characterization of the claudin-low intrinsic subtype of breast cancer. *Breast Cancer Res* **12**, 68 (2010).
18. Taube, J. H. *et al.* Core epithelial-to-mesenchymal transition interactome gene-expression signature is associated with claudin-low and metaplastic breast cancer subtypes. *Proc. Natl. Acad. Sci. U. S. A.* **107**, 15449–15454 (2010).
19. Finak, G. *et al.* MAST: a flexible statistical framework for assessing transcriptional changes and characterizing heterogeneity in single-cell RNA sequencing data. *Genome Biol.* **16**, 278 (2015).
20. Aibar, S. *et al.* SCENIC: Single-cell regulatory network inference and clustering. *Nat. Methods* **14**, 1083–1086 (2017).

6) Type of allocation and application for other funding

This application is for a UCBL grant.

An application will also be submitted for a doctoral grant from the « Ligue Nationale Contre le Cancer » (submission deadline: June 24, 2025).